Curriculum Vitae

Wookjin Choi, PhD · August 2026

Department of Radiation Oncology, Sidney Kimmel Medical College at Thomas Jefferson University, Philadelphia, PA 19107
Wookjin.Choi@jefferson.edu · https://qradiomics.com/

EDUCATION

- 2008-2013: Ph.D., Gwangju Institute of Science and Technology (Mechatronics, Medical Image Computing), Gwangju, South Korea
- 2006-2008: M.S., Gwangju Institute of Science and Technology (Mechatronics, Computer Vision), Gwangju, South Korea
- 2002-2006: B.S., Korea University of Technology and Education (Computer Science and Engineering), Cheonan, South Korea

POSTGRADUATE TRAINING AND FELLOWSHIP APPOINTMENTS

- 2018-2019: Department of Radiation Oncology (Research Associate), University of Virginia School of Medicine, Charlottesville, VA
- 2016-2018: Department of Medical Physics (Research Scholar), Memorial Sloan Kettering Cancer Center, New York, NY
- 2014-2016: Department of Radiation Oncology (Post-Doctoral Fellow), University of Maryland School of Medicine, Baltimore, MD
- 2013-2014: School of Mechatronics (Research Fellow), Gwangju Institute of Science and Technology, Gwangju, South Korea

MILITARY SERVICE

- 2008-2012: Technical Research Personnel (Alternative Military Service), Gwangju Institute of Science and Technology, Gwangju, South Korea

FACULTY APPOINTMENTS

- 2026-Present: Associate Professor, Department of Radiation Oncology, Sidney Kimmel Medical College at Thomas Jefferson University, Philadelphia, PA
- 2022-2026: Assistant Professor, Department of Radiation Oncology, Sidney Kimmel Medical College at Thomas Jefferson University, Philadelphia, PA
- 2019-2021: Assistant Professor, Department of Computer Science, Virginia State University, Petersburg, VA
- 2019-2021: Visiting Assistant Professor, Department of Radiation Oncology, University of Virginia School of Medicine, Charlottesville, VA

PROFESSIONAL / INDUSTRY ROLES

- 2013-2014: Founder and CEO, Qualia, Inc., Hwaseong, South Korea

AWARDS, HONORS, AND MEMBERSHIP IN HONORARY SOCIETIES

- 2026: 2026 ASTRO Annual Meeting Abstract Award, Basic Transitional Award in Physics (1 of 30 award-winning abstracts from more than 2,000 submissions)
- 2026: Selected Participant, NCI AI Embeddings Innovation Lab, “AI Embeddings for Next-Gen Cancer Diagnostics” (invitation-based, ~30 participants)
- 2024: Winner, NIH Biomedical Quantum Computation Challenge, Quantum Heart team. Awarded $25,000 as one of the top 3 teams in the $100K challenge.
- 2024: Invited Participant, NIH Quantum Computing Innovation Lab (international participation by 30 selected participants).
- 2021: Faculty of the Year Award, Korean American Scientists and Engineers Association (KSEA) Central Virginia Chapter
- 2020: Fourth Place Winner, Artificial Intelligence (AI) Tracks at Sea Challenge, hosted by Naval Information Warfare Center (NIWC) Pacific (Faculty Advisor)
- 2019: Travel Grant, KSEA SEED Workshop
- 2017: Travel Grant, Korean American Computer Scientists and Engineers Association (KOCSEA) Technical Symposium
- 2017: Bronze Medal (Top 5%), Data Science Bowl 2017 (Lung Cancer Screening) by Kaggle
- 2015: Top Performer (AUC 0.61 in a range of 0.50-0.68), LUNGx SPIE-AAPM-NCI Lung Nodule Classification Challenge
- 2015: Travel Grant, KSEA SEED Workshop
- 2014: Best Presentation Award, Korean Institute of Information Technology (KIIT)
- 2012: Best Paper Award, Korean Institute of Information, Electronics and Communication Technology (KIIECT)
- 2012: Second Place, Dasan Best Student Research Award, Gwangju Institute of Science and Technology (GIST)
- 2011: Best Presentation Award, KIIT
- 2008: Best Paper Award, KIIT
- 2008: Dasan Scholarship Recipient (outstanding academic performance, awarded upon Ph.D. admission), GIST
- 2006-2012: Korean Government Supported Full Scholarship (covering entire graduate tuition, stipend, and living expenses for M.S. and Ph.D. programs), GIST
- 2006-2012: Brain Korea 21 (BK21) Scholarship Recipient (M.S. and Ph.D. programs), GIST
- 2003-2005: BK21 Scholarship Recipient (undergraduate program), Korea University of Technology and Education (KoreaTech)
- 2003-2005: Internal Merit-based Scholarship (awarded throughout undergraduate degree based on merit), KoreaTech

MEMBERSHIPS IN PROFESSIONAL, SCHOLARLY, OR SCIENTIFIC SOCIETIES

- American Association of Physicists in Medicine (AAPM) - Member, 2014-Present
- American Society for Radiation Oncology (ASTRO) - Associate Member, 2023-Present; Member-in-Training, 2016-2019
- Institute of Electrical and Electronics Engineers (IEEE) - Member, 2010-Present
- Korean American Scientists and Engineers Association (KSEA) - Member, 2015-Present; CVA Chapter President, 2020-2021; Philadelphia Chapter President, 2025-Present
- Korean American Innovative Technology Engineers and Entrepreneurs (KITEE) - Member, 2023-Present; Executive Director, 2024-2025; Vice President, 2025-Present
- Korean American Medical Physicists in North America (KAMPiNA) - Member, 2015-Present
- Korean American Computer Scientists and Engineers Association (KOCSEA) - Member, 2015-Present

- Korean Institute of Information, Electronics and Communication Technology (KIIECT) - Member, 2010-2015
- Korean Institute of Information Technology (KIIT) - Member, 2008-2014

PROFESSIONAL, SCHOLARLY, OR SCIENTIFIC COMMITTEES

International

- 2017-2021: Member, Program Committee, International Workshop on Deep Learning in Bioinformatics, Biomedicine, and Healthcare Informatics (DLB2H 2017-2021), IEEE
- 2021: Chair, Local Organizing Committee, US-Korea Conference on Science, Technology and Entrepreneurship 2021, KSEA

National

- 2026: Co-Chair (Local Arrangement), STEP-UP 2026, KSEA
- 2025-Present: Councilor, National Council, KSEA
- 2025-Present: Vice President, Board of Directors, KITEE
- 2025-2026: Public Outreach Director, Directors Committee, KSEA
- 2025-2028: Secretary, Honor and Award Committee, KSEA
- 2024-2025: Executive Director, Board of Directors, KITEE
- 2021-2022: Councilor, National Council, KSEA

Regional

- 2026: Co-Chair, 2026 US-Korea Northeastern Symposium on AI, Bio-Medical, & Engineering, KSEA Philadelphia Chapter
- 2025-Present: President, Philadelphia Chapter, KSEA
- 2025, 2026: Chair & Founding Organizer, Prof. Chin OK Lee Award Evaluation Committee
- 2024, 2026: Chair, Scholarship Committee, KSEA Northeast Regional Conference (NRC) Scholarship
- 2023-2025: Member, Scholarship Committee, KSEA NRC Scholarship
- 2023, 2024: Conference Secretary, KSEA NRC 2023, 2024
- 2023-2026: Member, Steering Committee, KSEA NRC
- 2020: Judge, Senior Division Computer Science, Metro Richmond STEM Fair 2020

Institutional

- 2022-Present: Member, Jefferson Enterprise AI Center of Excellence Steering Committee
- 2026-Present: Member, AI@Jefferson Taskforce

MAJOR TEACHING ROLES AT THOMAS JEFFERSON UNIVERSITY

Teaching

1. 2023-Present: Radiomics for outcome and toxicity prediction and AI applications (didactic course for Radiation Oncology and Physics residents). Developed course and taught annually.
2. 2022: Self-paced course in Introduction to AI and Deep Learning. Developed course.

Advising / Mentoring

1. 2025: Isis Lloyd (Medical Student, Eastern Virginia Medical School at ODU), Research Mentor of Summer Research Internship
2. 2025: Moorin Khan (Medical Student, Drexel University), Research Mentor of Summer Research Internship

3. 2024-Present: Dr. Wenchao Cao (Physics Resident), Research Mentor
4. 2024-2025: Dr. Pradeep Bhetwal (Post-Doctoral Fellow), PI of Lung Cancer CBCT Radiomics Project
5. 2024-2025: Dr. Rupesh Ghimire (Post-Doctoral Fellow), PI of Interpretable Radiomics Project
6. 2024-2025: Michael Dichmann (Medical Student, Drexel University), Research Mentor of Summer Research Internship & PI of Lung Cancer CBCT Radiomics Project
7. 2024-2025: Dr. Ehsan Golkar (Post-Doctoral Fellow), Research Mentor
8. 2024: Lauren Nkwonta (Medical Student, Touro College of Osteopathic Medicine), Research Mentor
9. 2022-2023: Dr. Taindra Neupane (Post-Doctoral Fellow/Physics Resident), Research Mentor
10. 2022: Abhiram Kidambi (High School Student), Research Mentor of Summer Research Internship

MAJOR CLINICAL RESPONSIBILITIES

- 2022-Present: Computational Physicist, Department of Radiation Oncology, Thomas Jefferson University Hospital, Philadelphia, PA

INVITED EXTRAMURAL LECTURES

International

1. October 2024: “AI Research in Jefferson Health Radiation Oncology” - Reed Smith & KOAM Go-To-Market Digital Health Forum, New York, NY
2. November 2022: “Artificial Intelligence in Radiation Oncology” - Special Lecture, 2022 Annual Fall Conference of The Korean Society of Health Informatics and Statistics (Host: Dr. Chi Yeon Lim, Dongkuk University), Seoul, Korea
3. December 2021: “Artificial Intelligence in Radiation Oncology” - Yonsei University College of Medicine (Host: Dr. Jin Sung Kim, Yonsei University), Seoul, Korea
4. January 2018: “How to Effectively Operate Machine Learning Coursework for All Undergraduate Engineering Students” - Korea University of Technology and Education (KoreaTech) (Host: Dr. Oh-Young Kwon, KoreaTech), Cheonan, Korea

National

1. March 2022: “Artificial Intelligence in Radiation Oncology” - Mayo Clinic (Host: Dr. Tae Hyun Hwang, Mayo Clinic), Jacksonville, FL
2. October 2021: “Artificial Intelligence in Radiation Oncology” - Thomas Jefferson University (Host: Dr. Yevgeniy Vinogradskiy, TJU), Philadelphia, PA
3. November 2019: “Quantitative Cancer Image Analysis” - Memorial Sloan Kettering Cancer Center (Host: Dr. Joseph O. Deasy, MSKCC), New York, NY
4. September 2018: “Quantitative Image Analysis for Cancer Diagnosis and Radiation Therapy” - University of Virginia School of Medicine (Host: Dr. Jeffrey Siebers, UVA), Charlottesville, VA
5. May 2018: “Quantitative Image Analysis for Cancer Diagnosis and Radiation Therapy” - Vanderbilt University Medical Center (Host: Dr. Michael Price, Vanderbilt University), Nashville, TN

Local

1. September 2025: “Transforming Radiation Oncology with Artificial Intelligence” - Philadelphia Korean Scholars Association (PKSA), Philadelphia, PA
2. April 2024: “Transforming Radiation Oncology: The Clinical Impact of Artificial Intelligence and the Role of Computational Medical Physicists” - Data/Networks/AI of KSEA NRC 2024 (Host: Dr. Heeyoung Kwon, Meta), Montclair, NJ

3. January 2020: “Artificial Intelligence in Radiation Oncology” - KSEA CVA Seminar (Host: Dr. B. Brian Park, UVA), Charlottesville, VA

NOTABLE INTRAMURAL LECTURES / WORKSHOPS

1. May 2025: “Bridging AI and Quantum Computing for Precision Radiotherapy” - Medical Physics Mini-Series: Computational and AI Advancement in Radiation Oncology
2. October 2024: “External Beam Plan Check Automation” - Enterprise Medical Physics Division Retreat
3. October 2023: “Computational Program at Jefferson” - Enterprise Medical Physics Division Retreat
4. May 2022: “Artificial Intelligence in Magnetic Resonance Guided Radiotherapy” - MRgRT 2022 Inaugural Retreat

BIBLIOGRAPHY

Publications, Peer Reviewed

Peer-Reviewed Journal Articles

1. Cao W, Lloyd I, Dichmann M, Haldar N, Thomas D, Chen Z, Blomain E, et al., Choi W. (2026). Cross-institutional validation of a novel LLM-based cardiac event extraction framework from electronic health records. International Journal of Radiation Oncology, Biology, Physics. (In press)
2. Haldar N, Okazaki K, McLean LT, Mourtada F, Taleei R, Li J, Caster J, Anne PR, Wan S, Vinogradskiy Y, Mooney KE, Choi W, Chen Y. (2026). Contour and dosimetric evaluation of an in-room mobile CBCT scanner for cylinder based brachytherapy. Brachytherapy.
3. Neupane T, Castillo E, Chen Y, Pahlavian SH, Castillo R, Vinogradskiy Y, Choi W. (2025). Predicting radiation pneumonitis in lung cancer patients using robust 4DCT-ventilation and perfusion imaging. Radiotherapy and Oncology 212, 111110.
4. Ciaccio P, Lombardo J, Fuquay A, Pahlavian S, Grimm R, Kang J, Choi W, Sullivan P, Vinogradskiy Y. (2025). Development and validation of an AI-based lung lobe auto-contouring tool using radiation therapy planning free-breathing images. Reports of Practical Oncology and Radiotherapy 30(6), 767-772.
5. Logan JA, Sadhu S, Hazlewood C, Denton M, Burke SE, Simone-Soule CA, Black C, Ciaverelli C, Stulb J, Nourzadeh H, Vinogradskiy Y, Leader A, Dicker AP, Choi W, Simone NL. (2025). Bridging gaps in cancer care: Utilizing large language models for accessible dietary recommendations. Nutrients 17(7), 1176.
6. Li J, Choi W, Anne R. (2025). Deep learning-based auto-segmentation for liver Yttrium-90 selective internal radiation therapy. Technology in Cancer Research & Treatment 24, 15330338251327081.
7. Lee M, Kim JH, Choi W, Lee KH. (2025). AI-assisted segmentation tool for brain tumor MR image analysis. Journal of Imaging Informatics in Medicine 38(1), 74-83.
8. Choi W, Jia Y, Kwak J, Werner-Wasik M, Dicker AP, Simone NL, Storozynsky E, Jain V, Vinogradskiy Y. (2024). Novel functional radiomics for prediction of cardiac positron emission tomography avidity in lung cancer radiotherapy. JCO Clinical Cancer Informatics 8, e2300241. (Featured in JCO CCI Editorial)
9. Lu W, Hong LX, Yamada N, Berry SL, Song Y, Choi W, Cervino LI, Tang X, Mechalakos JG, Romesser PB, Powell S, Li G. (2023). Comparison of setup accuracy of optical surface image versus orthogonal x-ray images for VMAT of the left breast using deep-inspiration breath-hold. Journal of Applied Clinical Medical Physics 24, e14117.
10. Choi W, Liu CJ, Riyahi Alam S, Oh JH, Vaghjiani R, Humm J, Weber W, Adusumilli PS, Deasy JO, Lu W. (2023). Preoperative 18F-FDG PET/CT and CT radiomics for identifying aggressive histopathological subtypes in early stage lung adenocarcinoma. Computational and Structural Biotechnology Journal 21, 5601-5608.
11. Wang J, Choi W, Diaz J, Trott C. (2022). The 3D position estimation and tracking of a surface vehicle using a mono-camera and machine learning. Electronics 11(14), 2141.

12. Oh JH*, Choi W*, Ko E, Kang M, Tannenbaum A, Deasy JO. (2021). PathCNN: Interpretable convolutional neural networks for survival prediction and pathway analysis applied to glioblastoma. Bioinformatics 37(Supplement_1), i443-i450. (*Joint First Authors)
13. Choi W, Nadeem S, Riyahi S, Deasy JO, Tannenbaum A, Lu W. (2021). Reproducible and interpretable spiculation quantification for lung cancer screening. Computer Methods and Programs in Biomedicine 200, 105839.
14. Alves VGL, Aliotta E, Ahmed M, Siebers JV. (2021). An error detection method for real-time EPID-based treatment delivery quality assurance. Medical Physics 48(3), 1335-1345.
15. Aliotta E, Nourzadeh H, Choi W, Alves VGL, Siebers JV. (2020). An automated workflow to improve efficiency in radiation therapy treatment planning by prioritizing organs-at-risk. Advances in Radiation Oncology 5(6), 1324-1333.
16. Garau N, Paganelli C, Summers P, Choi W, Alam S, Lu W, Fanciullo C, Bellomi M, Baroni G, Rampinelli C. (2020). External validation of radiomics-based predictive models in low-dose CT screening for early lung cancer diagnosis. Medical Physics 47(9), 4044-4053.
17. Wang J, Zhang H, Chuong M, Latifi K, Tan S, Choi W, Hoffe S, Shridhar R, Lu W. (2019). Prediction of anal cancer recurrence after chemoradiotherapy using quantitative image features extracted from serial 18F-FDG PET/CT. Frontiers in Oncology 9, 934.
18. Zhang J, Markova S, Garcia A, Huang K, Nie X, Choi W, Lu W, Wu A, Rimner A. (2018). Potential clinical utility of automatic contour propagation in T2-weighted navigator-triggered 4DMRI for internal organ-at-risk volume (IOV) evaluation. Journal of Applied Clinical Medical Physics 19(5), 308-316.
19. Choi W, Riyahi S, Kligerman SJ, Liu CJ, Mechalakos JG, Lu W. (2018). Technical note: Identification of normal lung CT texture features robust to tumor size for the prediction of radiation-induced lung disease. International Journal of Medical Physics, Clinical Engineering and Radiation Oncology 7(3), 330-338.
20. Riyahi S, Choi W, Liu CJ, Zhong H, Wu AJ, Mechalakos JG, Lu W. (2018). Quantifying local tumor morphological changes with Jacobian map for prediction of pathologic tumor response to chemo-radiotherapy in locally advanced esophageal cancer. Physics in Medicine & Biology 63(14), 145020.
21. Choi W, Oh JH, Riyahi S, Jiang F, Chen W, White C, Rimner A, Mechalakos JG, Deasy JO, Lu W. (2018). Radiomics analysis of pulmonary nodules in low-dose CT for early detection of lung cancer. Medical Physics 45(4), 1537-1549. (Editor's Pick)
22. Tan S, Li L, Choi W, Kang MK, D'Souza W, Lu W. (2017). Adaptive region-growing with maximum curvature strategy for tumor segmentation in 18F-FDG PET. Physics in Medicine and Biology 62(13), 5383-5402.
23. Choi W, Xue M, Lane B, Kang M, Patel K, Regine W, Klahr P, Wang J, Chen S, D'Souza W, Lu W. (2016). Individually optimized contrast-enhanced 4D-CT for radiotherapy simulation in pancreatic ductal adenocarcinoma. Medical Physics 43(10), 5659-5666.
24. Jaffar MA, Choi W, Choi TS. (2014). Classification of lungs nodules using hybrid features and neural network. Information 17(5), 1771-1776.
25. Choi W, Choi TS. (2014). Automated pulmonary nodule detection based on three-dimensional shape-based feature descriptor. Computer Methods and Programs in Biomedicine 113(1), 37-54.
26. Choi W, Choi TS. (2013). Log-polar sampling based voxel classification for pulmonary nodule detection in lung CT scans. Journal of Korean Institute of Information, Electronics and Communication Technology 6(1), 37-44.
27. Choi W, Choi TS. (2013). Automated pulmonary nodule detection system in computed tomography images: A hierarchical block classification approach. Entropy 15(2), 507-523.
28. Choi W, Choi TS. (2012). Pulmonary nodule detection based on hierarchical 3D block analysis in chest CT scans. Journal of Korean Institute of Information, Electronics and Communication Technology 5(1), 13-19.
29. Choi W, Choi TS. (2012). Genetic programming-based feature transform and classification for the automatic detection of pulmonary nodules on computed tomography images. Information Sciences 212, 57-78.
30. Choi W, Muhammad MS, Choi TS. (2008). 3D shape recovery algorithm with reduction of 3D spatial complexity. Journal of Korean Institute of Information Technology 6(4), 108-114.
31. Mahmood MT, Choi W, Choi TS. (2008). PCA-based method for 3-D shape recovery of microscopic objects from image focus using discrete cosine transform. Microscopy Research and Technique 71(12), 897-907.

Peer-Reviewed Conference Papers

32. Lee S, Zhao Y, Choi W. (2024, July). High-rate emphasized DeepLabV3Plus for semantic segmentation of breast cancer-related hematoxylin and eosin-stained images. Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 1-4. Orlando, FL, USA.
33. Choi W, Dahiya N, Nadeem S. (2022). CIRDataset: A large-scale dataset for clinically-interpretable lung nodule radiomics and malignancy prediction. Medical Image Computing and Computer Assisted Intervention (MICCAI) 2022. Lecture Notes in Computer Science, vol 13438. Springer, Cham.
34. Lee S, Farley C, Shim S, Zhao Y, Choi W, Yoo WS. (2020). Unsupervised learning of deep-learned features from breast cancer images. 2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE), 740-745.
35. Gladman D, Osegbe J, Choi W*, Lee JS. (2020). Automatic motion tracking system for analysis of insect behavior. Proc. SPIE 11510, Applications of Digital Image Processing XLIII, 115102W. (*Corresponding author)
36. Choi W, Nadeem S, Riyahi S, Deasy JO, Tannenbaum A, Lu W. (2018). Interpretable spiculation quantification for lung cancer screening. Shape in Medical Imaging: International Workshop (ShapeMI 2018). Lecture Notes in Computer Science 11167, 38-48.
37. Riyahi S, Choi W, Liu CJ, Nadeem S, Tan S, Zhong H, Chen W, Wu A, Mechalakos J, Deasy J, Lu W. (2018). Quantification of local metabolic tumor volume changes by registering blended PET-CT images for prediction of pathologic tumor response. Data Driven Treatment Response Assessment and Preterm, Perinatal, and Paediatric Image Analysis (PIPPI 2018, DATRA 2018). Lecture Notes in Computer Science 11076, 31-41.
38. Choi W, Choi TS. (2014, May). Pulmonary nodule detection using Insight Toolkit in lung CT images. Korean Institute of Information Technology (KIIT) Conference, 9-13.
39. Choi W, Choi TS. (2012, November). Statistical shape model based three dimensional shape modeling for knee joint on magnetic resonance images. KIIT Conference, 461-465.
40. Choi W, Choi TS. (2012, May). Pulmonary nodule detection using voxel classification in lung CT scans. Korean Institute of Information, Electronics and Communication Technology (KIIECT) Conference, 152-155.
41. Choi W, Choi TS. (2011, October). Pulmonary structures segmentation and nodule detection using 3D block analysis of chest CT scans. KIIECT Conference, 197-200.
42. Choi W, Choi TS. (2011, May). Pulmonary nodule candidate detection using modified circular Hough transform. KIIT Conference, 8-11.
43. Mahmood MT, Lee IH, Choi W, Choi TS. (2011, January). A nonlinear approach for depth from focus for digital cameras. IEEE International Conference on Consumer Electronics (ICCE), 187-188.
44. Choi W, Choi TS. (2010). Pulmonary nodule detection using genetic programming. KIIECT Conference 3(2), 100-103.
45. Choi W, Choi TS. (2010, September). Computer-aided detection of pulmonary nodules using genetic programming. IEEE International Conference on Image Processing (ICIP), 4353-4356.
46. Choi W, Lee IH, Choi TS. (2010, May). Adaptive lung region segmentation using graph-cuts. KIIT Conference, 7-10.
47. Choi W, Choi TS. (2009, June). False positive reduction in pulmonary nodule detection using principal component analysis. KIIT Conference, 202-207.
48. Choi W, Choi TS. (2009). Computerized detection of pulmonary nodule based on two-dimensional PCA. International Conference on Computational Science and Its Applications (ICCSA) 2009. Lecture Notes in Computer Science 5593, 693-702. [ORAL]
49. Choi W, Majid A, Choi TS. (2009). False positive reduction for pulmonary nodule detection using two-dimensional principal component analysis. Proc. SPIE 7443, Applications of Digital Image Processing XXXII, 744322.
50. Mahmood MT, Choi W, Choi TS. (2008). DCT and PCA based method for shape from focus. ICCSA 2008. Lecture Notes in Computer Science 5073, 1025-1103. [ORAL]
51. Choi W, Choi TS. (2008). Fast three-dimensional shape recovery in TFT-LCD manufacturing. Proc. SPIE 7073, Applications of Digital Image Processing XXXI, 70731U.

52. Choi W, Muhammad MS, Lee M, Choi TS. (2008, June). Depth measurement using pixel intensities. Institute of Electronics Engineers of Korea (IEEK) Conference, 901-902.

Abstracts

Oral Presentations (Presenting Author)

1. Choi W, Bhetwal P, Dichmann M, Jia Y, Cao W, Liang D, Chen Y, Dicker A, Vinogradskiy Y. (2026, September). Early adaptive interventions in lung cancer: Leveraging fusion of longitudinal CBCT trajectories and clinical variables for robust survival prediction. ASTRO Annual Meeting 2026. Abstract #: 75557. [BEST of Physics Oral; 2026 ASTRO Annual Meeting Abstract Award, Basic Transitional Award in Physics]
2. Choi W, Bhetwal P, Dichmann M, Jia Y, Cao W, Chen Y, Vinogradskiy Y, Nourzadeh H. (2026, July). Hierarchical spatial and temporal aggregation as progressive denoising for longitudinal CBCT radiomics in lung cancer. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: SU-315-GPD-T-25. [SNAP Oral]
3. Choi W, Jia Y, Wang W, Vinogradskiy Y. (2024). Novel AI-based dose prediction directly from diagnostic PET/CT: Applications for multi-disciplinary lung cancer care. AAPM 66th Annual Meeting & Exhibition 2024. [Snap Oral]
4. Choi W, Dicker AP, Vinogradskiy Y. (2023). Artificial intelligence to reduce radiation-induced cardiotoxicity in lung cancer radiotherapy: A novel functional radiomics using cardiac FDG-PET/CT. INFORMS Annual Meeting 2023. [Oral]
5. Choi W, Jia Y, Kwak J, Dicker AP, Simone NL, Storozynsky E, Jain V, Vinogradskiy Y. (2023). Novel functional radiomics for prediction of cardiac PET avidity in lung cancer radiotherapy. ASTRO Annual Meeting 2023. [Quick Pitch Oral]
6. Choi W, Aliotta E, Nourzadeh H, Siebers J. (2019). Simulation of realistic organ-at-risk delineation variability in head and neck radiation therapy. Medical Physics 46(6), E186-E187. AAPM Annual Meeting 2019. [Snap Oral]
7. Choi W, Oh J, Riyahi Alam S, Jiang F, Chen W, Deasy J, Lu W. (2017). Radiomics analysis of pulmonary nodules in low dose CT for early detection of lung cancer. AAPM Annual Meeting 2017. [ORAL]
8. Choi W, Liu C, Riyahi Alam S, Oh J, Adusumilli P, Weber W, Deasy J, Lu W. (2017). Aggressive lung adenocarcinoma subtype prediction using FDG-PET/CT radiomics. AAPM Annual Meeting 2017. [Snap Oral]
9. Choi W, Xue M, Lane B, Kang M, Patel K, Regine W, Klahr P, Wang J, Chen S, D'Souza W, Lu W. (2016). Individually optimized contrast-enhanced 4D-CT for radiotherapy simulation in pancreatic ductal adenocarcinoma. AAPM Annual Meeting 2016. [Oral]
10. Choi W, Xue M, Lane B, Kang M, Patel K, Regine W, Klahr P, Wang J, Chen S, D'Souza W, Lu W. (2016). Individually optimized contrast-enhanced 4D-CT for radiotherapy simulation in pancreatic cancer. ICCR 2016. [Oral]

Oral Presentations (Mentored/Trainee-Led)

11. Bhetwal P, Dichmann M, Ghimire R, Chen Y, Vinogradskiy Y, Werner-Wasik M, Dicker A, Choi W. (2025, July). Development and validation of a scalable radiomics pipeline for lung cancer research using clinical and public datasets. AAPM 67th Annual Meeting & Exhibition 2025. Medical Physics 52(10), e700597. Final Paper #: SU-1015-202-4. [Oral]
12. Dichmann M, Haldar N, Storozynsky E, Jia Y, Simone NL, Werner-Wasik M, Dicker A, Vinogradskiy Y, Choi W. (2025, July). PET imaging and novel cardiac radiomics to predict pre-radiotherapy cardiac conditions for lung cancer patients undergoing radiotherapy. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: TU-1015-209-3. [Snap Oral]
13. Cao W, Dichmann M, Haldar N, Storozynsky E, Jia Y, Simone N, Werner-Wasik M, Dicker AP, Vinogradskiy Y, Choi W. (2025, October). Novel cardiac radiomics using PET imaging and deep learning to predict post-radiotherapy cardiac toxicity for lung cancer patients. International Journal of Radiation Oncology, Biology, Physics 123(1), S253-S254. ASTRO Annual Meeting 2025. Abstract #: 67661. [Quick Pitch Oral]

14. Neupane T, Castillo E, Chen Y, Pahlavian SH, Castillo R, Vinogradskiy Y, Choi W. (2024). Predicting radiation pneumonitis with robust 4DCT-ventilation and 4DCT-perfusion imaging using prospective lung cancer clinical trial data. International Journal of Radiation Oncology, Biology, Physics 120(2), S141. ASTRO Annual Meeting 2024. [Best of Physics]

Collaborator Oral Presentations

15. Haldar N, Xiao Y, Wen D, Choi W, Vinogradskiy Y, Bradley J, Hu C, Werner-Wasik M. (2025, September). Impact of AI-based coronary artery calcium (CAC) scores and heart dose on survival in locally-advanced non-small-cell lung cancer (LA-NSCLC): A secondary analysis of RTOG 0617. International Journal of Radiation Oncology, Biology, Physics 123(1), S160-S161. ASTRO Annual Meeting 2025. Abstract #: 68016. [Oral]
16. Cao W, Mourtada F, Choi W, Vinogradskiy Y, Anne R, Chen Y. (2025, July). A self-supervised deep learning approach for automatic identification and metal artifact reduction in cone-beam CT for brachytherapy. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: WE-305-202-5. [Oral]
17. Haldar N, Taleei R, Mourtada F, Li J, Anne P, Wan S, Vinogradskiy Y, Mooney KE, Choi W, Chen Y. (2024). Clinical validation of using an in-room mobile CBCT scanner for cylinder-based brachytherapy planning. Brachytherapy 23(6), S121-S122.
18. Nourzadeh H, Ahmed M, Choi W, Aliotta E, Watkins W, Siebers J. (2019). Radiation Therapy Robustness Analyzer (RTRA): A GPU accelerated software to simulate the effect of uncertainties in external beam radiation therapy. Medical Physics 46(6), E415-E415. AAPM Annual Meeting 2019. [Oral]
19. Riyahi S, Choi W, Liu C, Nadeem S, Tan S, Zhong H, Chen W, Wu A, Mechalakos J, Deasy J, Lu W. (2019). Quantification of local metabolic tumor volume changes by registering blended 18F-FDG PET/CT images for prediction of pathologic tumor response. Medical Physics 46(6), E463-E463. AAPM Annual Meeting 2019. [Oral]

Poster Presentations (Presenting Author)

20. Choi W, Nourzadeh H, Thomas DH, Chen Y, Vinogradskiy Y. (2026, July). Multi-tier automated pipeline for TG-263 structure name standardization using rule-based, NLP, and RAG-enhanced LLM matching. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: SU-115-220-8.
21. Choi W, Nourzadeh H, Chen Y, Ainsley CG, Desai VK, Kubli AA, Vinogradskiy Y, Mooney KE, Werner-Wasik M, Mueller A. (2023). Novel deep learning segmentation models for accurate GTV and OAR segmentation in MR-guided adaptive radiotherapy for pancreatic cancer patients. ASTRO Annual Meeting 2023.
22. Choi W, Vinogradskiy Y. (2023). Novel functional delta-radiomics for predicting overall survival in lung cancer radiotherapy using cardiac FDG-PET uptake. AAPM Annual Meeting 2023.
23. Choi W, Nourzadeh H, Chen Y, Ainsley CG, Desai VK, Kubli AA, Vinogradskiy Y, Werner-Wasik M, Mueller A, Mooney KE. (2023). Deep learning segmentation for accurate GTV and OAR segmentation in MR-guided adaptive radiotherapy for pancreatic cancer patients. AAPM Annual Meeting 2023.
24. Choi W, Nourzadeh H, Lee S, Chen Y, Ghassemi N, Dicker A, Vinogradskiy Y. (2022). A framework for reproducible and scalable radiomics pipelines for radiation oncology using self-contained containers and workflow manager. Medical Physics 49(6), E190-E191. AAPM Annual Meeting 2022. [Interactive ePoster]
25. Choi W, Leandro Alves V, Aliotta E, Nourzadeh H, Siebers J. (2020). Assessing the dosimetric links between organ-at-risk delineation variability and treatment planning variability. Joint AAPM | COMP Meeting 2020. [ePoster]
26. Choi W, Riyahi S, Liu CJ, Lu W. (2017). Robust normal lung CT texture features for the prediction of radiation-induced lung disease. ASTRO Annual Meeting 2017. [ePoster]
27. Choi W, Xue M, Lane B, Kang M, Patel K, Regine W, Klahr P, Wang J, Chen S, D'Souza W, Lu W. (2016). Individually optimized contrast-enhanced 4D-CT for radiotherapy simulation in pancreatic adenocarcinoma. ASTRO Annual Meeting 2016. [Selected for the ARRO poster walk]
28. Choi W, Riyahi S, Lu W. (2016). Identification of robust normal lung CT texture features for the prediction of radiation-induced lung disease. AAPM Annual Meeting 2016.

29. Choi W, Xue M, Kang MK, Patel K, Regine W, Klahr P, Wang J, D'Souza W, Lu W. (2015). Image quality assessment of contrast-enhanced 4D-CT for pancreatic adenocarcinoma in radiotherapy simulation. AAPM Annual Meeting 2015.
30. Choi W, Kang MK, Wang J, Lu W. (2015). Quantitative image feature analysis of multiphase liver CT for hepatocellular carcinoma (HCC) in radiation therapy. AAPM Annual Meeting 2015.
31. Choi W, Muhammad MS, Choi TS. (2008, May). 3D space complexity reducing method for 3D camera. KIIT Conference 2008.
32. Choi W, Choi TS. (2008, May). Three-dimensional shape recovery for measuring the protrusion on color filter. KIIT Conference 2008.
33. Choi W, Choi TS. (2008, January). Three-dimensional shape recovery for measuring LCD color filter. Workshop on Image Processing and Image Understanding 2008.
34. Choi W, Eom S, Moon H, Choi TS. (2007, October). Daphnia tracking algorithm for monitoring water quality. Korea Signal Processing Conference.

Poster Presentations (Mentored/Trainee-Led)

35. Cao W, Panchal D, Haldar N, Dichmann M, Faherty P, Blomain E, Choi W, Vinogradskiy Y. (2026, July). Improving cardiac toxicity prediction from PET radiomics: Cross-cohort validation and late-fusion ensemble modeling. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: SU-315-GPD-R-23.
36. Panchal D, Aragam V, Kove CM, Marhefka S, Thomas DH, Vinogradskiy Y, Choi W. (2026, July). Novel constraint-based method to evaluate and score patient radiotherapy plans. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: PO-DPV-T-272.
37. Cao W, Choi W, Vinogradskiy Y. (2026, July). Novel LLM-based cardiotoxicity extraction from electronic health records. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: SU-315-GPD-T-630.
38. Ghimire R, Bhetwal P, Nkwonta L, Wang W, Jia Y, Dicker A, Werner-Wasik M, Vinogradskiy Y, Choi W. (2025, July). AI-based SBRT dose prediction directly from diagnostic PET/CT: Applications for multi-disciplinary lung cancer care. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: SU-330-GPD-T-128.
39. Bhetwal P, Nourzadeh H, Chen Y, Desai VK, Ainsley CG, Vinogradskiy Y, Werner-Wasik M, Mueller A, Mooney KE, Choi W. (2025, July). Evaluating deep learning models for accurate segmentation of GTV and OARs in MR-guided adaptive radiotherapy for pancreatic cancer. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: SU-330-GPD-T-216.
40. Bhetwal P, Dichmann M, Ghimire R, Chen Y, Vinogradskiy Y, Werner-Wasik M, Dicker AP, Choi W. (2025, September). Integrating clinical and radiomic features for enhanced prognostic modeling for lung cancer survival. International Journal of Radiation Oncology, Biology, Physics 123(1), e719. ASTRO Annual Meeting 2025. Abstract #: 68241. [Poster]

Collaborator Poster Presentations

41. Li J, Vinogradskiy Y, Choi W. (2026, July). Auto-segmentation of metastatic liver tumors in SIRT. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: PO-DPV-R-6.
42. Nourzadeh H, Thomas DH, Desai VK, Cao W, Choi W. (2026, July). Automated MPPG5b-compliant platform for photon beam model validation across an integrated health care system. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: SU-315-GPD-T-141.
43. Nourzadeh H, Choi W, Mund KW, Kang J, Fu L, Mourtada F, Dicker A, Vinogradskiy Y. (2026, July). A vendor-agnostic radiation treatment plan quality assurance software implemented for an integrated health system. 2026 Joint AAPM|COMP Annual Meeting & Exhibition. Final Paper #: SU-315-GPD-T-1025.
44. Ghassemi N, Russial O, Choi W, Vinogradskiy Y, Li J. (2025, July). Evaluation of MIM Contour ProtegeAI for auto-contouring organs at risk in radiation therapy: A multi-site retrospective study. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: SU-330-GPD-T-583.

45. Thomas DH, Vinogradskiy Y, Choi W, Romanofski D, Lamb JM. (2025, July). Intelligent black box recording for radiation therapy: Feasibility study of vision-language models for treatment monitoring. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: SU-330-GPD-T-269.
46. Li J, Choi W. (2025, July). Evaluate a deep-learning auto-segmentation software for liver SIRT. AAPM 67th Annual Meeting & Exhibition 2025. Final Paper #: SU-330-GPD-R-44.
47. Nourzadeh H, Choi W, Vinogradskiy Y, Chen Y, Yu Y, Shi W, Liu H. (2024). Deep learning model incorporating geometric features for multi-metastatic Gamma Knife stereotactic radiosurgery dose distribution. AAPM 66th Annual Meeting & Exhibition 2024.
48. Li J, Choi W, Vinogradskiy Y. (2024). Validation of MIM Contour Protege AI auto-contouring for radiotherapy. AAPM 66th Annual Meeting & Exhibition 2024.
49. Mooney KE, Belko SA, Williams J, Gerry A, Choi W, Daley M, Pugliese RS. (2023). An MR-conditional motor for 1D water tank measurements in an MR-Linac. AAPM Annual Meeting 2023.
50. Rashad H, Karki A, Czak J, Alves VGL, Nourzadeh H, Choi W, Siebers JV. (2023). Quantitative assessment of dosimetric effect of using alternative spinal cord delineation in treatment planning as functions of delineations, setup uncertainty and planning techniques using an alternative truth assessment method. AAPM Annual Meeting 2023.
51. Rashad M, Badry F, Alves V, Nourzadeh H, Choi W, Siebers J. (2022). Clinical equivalency of alternate head-and-neck OAR delineations. Medical Physics 49(6), E750-E750. AAPM Annual Meeting 2022. [ePoster]
52. Lu W, Yamada N, Hong L, Choi W, Tang X, Mechalakos J, Berry S, Romesser P, Cahlon O, Powell S, Li G. (2020). Comparison of surface imaging and 2DkV imaging vs. CBCT in left breast DIBH radiotherapy. Joint AAPM | COMP Meeting 2020. [ePoster]
53. Soomro MH, Nourzadeh H, Leandro Alves V, Choi W, Siebers J. (2020). Dosimetric analysis of OARnet auto-delineations for head and neck organs-at-risk. Joint AAPM | COMP Meeting 2020. [ePoster]
54. Thor M, Fitzgerald KJ, Apte A, Oh JH, Iyer A, Nadeem S, Choi W, Yorke ED, Chaft J, Wu AJ, Offin M, Gelblum D, Deasy JO, Rimner A. (2020). A pre-treatment PET and CT-derived model for progression-free survival in early stage non-small cell lung cancer. ASTRO Annual Meeting 2020.
55. Riyahi S, Choi W, Liu CJ, Lu W. (2017). Prediction of pathologic tumor response to chemoradiotherapy in locally advanced esophageal cancer using Jacobian map. ASTRO Annual Meeting 2017.
56. Li G, Markova S, Garcia A, Moody J, Zhang J, Rimner A, Wu A, Crane C, Choi W, Lu W, Tyagi N, Zakian K, Kadbi M, Mechalakos J, Deasy J. (2017). Clinical evaluation of automatic organ contour propagation using 4DMRI images. ASTRO Annual Meeting 2017.
57. Riyahi Alam S, Choi W, Liu C, Lu W. (2017). Quantifying tumor morphological change with Jacobian map for prediction of pathologic tumor response to chemo-radiotherapy in locally advanced esophageal cancer. AAPM Annual Meeting 2017. [ePoster]
58. Zhang H, Riyahi Alam S, Choi W, Ayala-Peacock D, McTyre E, Bourland J, Blackstock A, D’Souza W, Lu W. (2017). First experience on validation of pathologic response prediction models utilizing PET/CT radiomic features using multi-institutional datasets. AAPM Annual Meeting 2017. [ePoster]
59. Riyahi S, Choi W, Bhooshan N, Tan S, Zhang H, Lu W. (2016). Comparison of registration methods for modeling pathologic response of esophageal cancer to chemoradiation therapy. AAPM Annual Meeting 2016. [Snap Oral]
60. Riyahi S, Choi W, Lu W. (2016). Novel radiomics quantifying tumor structural evolution using deformation vector field: Application for tumor response assessment. AAPM Annual Meeting 2016.
61. Chuong MD, Zhang HH, Wang J, Latifi K, Saeed N, Tan S, Choi W, Hoffe SE, Shridhar R, Lu W. (2015). Analytics for progression free survival and distant metastasis prediction of anal cancer patients after chemoradiation therapy using spatial temporal FDG-PET/CT features. ASTRO Annual Meeting 2015.
62. Wang J, Chuong M, Latifi K, Saeed N, Tan S, Choi W, Hoffe S, Shridhar R, Moros E, Lu W. (2015). Repeated 18F-FDG PET/CTs based feature analysis for the prediction of anal cancer recurrence. AAPM Annual Meeting 2015. [SNAP ORAL]

63. Zhang H, Wang J, Chuong M, D’Souza W, Latifi K, Saeed N, Tan S, Choi W, Hoffe S, Shridhar R, Moros E, Lu W. (2015). Evaluating the role of mid-treatment and post-treatment FDG-PET/CT in predicting progression-free survival and distant metastasis. AAPM Annual Meeting 2015.
64. Zhang HH, Wang J, Chuong M, Latifi K, Tan S, Choi W, Hoffe S, Shridhar R, Lu W. (2015). Analytics in predicting progression-free survival and distant metastasis after chemoradiotherapy using spatial-temporal FDG-PET/CT features. SNMMI Annual Meeting 2015.
65. Jaffar MA, Choi W, Choi TS. (2012, March). Classification of pulmonary nodule from low dose CT scan images using neural networks. Advanced Signal Processing 2012.

Editorials, Reviews, Chapters, Including Participation in Committee Reports

1. Kim J, Choi W. (2014). Radiomics: Novel paradigm of deep learning for clinical decision support toward Plan B using liquid biopsy. Korean Association for Radiation Application Issue Paper (04).

ISSUED PATENTS AND OTHER INTELLECTUAL PROPERTY

1. 2016: Xue M, Lu W, Zhang H, Kligerman S, D’Souza W, Choi W. U.S. Patent Number 9,456,798 B2, Methods and Systems for Individually Optimizing Uniform Contrast Enhancement in Computed Tomography Imaging.
2. 2016: Choi TS, Choi W. U.S. Patent Number 9,349,074 B2, Method and Apparatus for Generating 3-D Knee Joint Image.
3. 2016: Choi TS, Choi W. Korea Patent Number 10-1594994, Method and Apparatus for Generating 3-D Knee Joint Image.
4. 2011: Choi TS, Choi W. Korea Patent Number 10-1085949, Apparatus for Classifying a Lung and a Method Thereof, Capable of Automatically Determining the State of a Lung.
5. 2011: Choi TS, Choi W, Malik AS. Korea Patent Number 10-1066468, Method and Device for Discriminating a Honeycombed Lung from a Normal Lung and a Computer Readable Recording Medium.

Invention Disclosures

6. 2025: Cao W, Vinogradskiy Y, Choi W. Two-Phase Reasoning for Automated Cardiac Event Recognition (TRACER). A chain-of-thought prompting framework optimized for cardiac event extraction from clinical data using a two-phase pipeline (structured data analysis and LLM-based context-aware analysis).
7. 2025: Vinogradskiy Y, Dicker A, Choi W. Radiomics-Driven Clinical Classification of Cardiac Sarcoidosis Using Myocardial FDG PET Imaging. An AI model using machine learning algorithms (TPOT, XGBoost) to predict cardiac sarcoidosis by analyzing myocardial FDG PET scans.
8. 2025: Vinogradskiy Y, Lamb J, Pijanowski J, Choi W. A Method and System for a Relational Back-Up System and Minimum Functionality Algorithms for Radiation Oncology Ransomware Attack Resiliency. A relational backup system (RBS) designed to maintain critical patient treatment data and functionality during radiation oncology ransomware attacks.
9. 2024: Nourzadeh H, Choi W. Prototype Clinical Software for Distributed Data Systems. Clinical software prototype optimized for distributed data environments in radiation oncology settings.
10. 2023: Vinogradskiy Y, Dicker A, Choi W. A Method for Radiation Dose Prediction Using Diagnostic Imaging. AI/ML-based approach to predict radiation dose distributions directly from diagnostic imaging without manual delineation.
11. 2023: Vinogradskiy Y, Dicker A, Choi W. A Method for PET-Based Functional Radiomics for Prediction of Cardiotoxicity for Cancer Patients. Computational method extracting radiomic features from cardiac FDG PET-CT to predict cardiotoxicity and support clinical decision-making.

COMPANIES STARTED OR OTHER COMMERCIALIZATION OF RESEARCH / INVENTIONS

2013-2014: QuaLIA, Inc.

- A startup founded to commercialize patented Ph.D. research, awarded a competitive grant from the Small and Medium Business Administration of Korea (a program analogous to the U.S. SBIR).

GRANT / FUNDING ACTIVITIES INCLUDING CLINICAL TRIALS

Present

1. Choi, W (PI). Longitudinal CBCT radiomics analysis for lung cancer radiotherapy response and prognosis prediction. Varian Medical Systems Research Grant (12/2023-12/2026). Total Cost: $230,000.
2. Vinogradskiy, Y (PI), Choi, W (Co-I). Systems Engineering for Ransomware Attack Response Resiliency in Radiation Oncology. NIH Agency for Healthcare Research and Quality (AHRQ), R18HS030060 (08/2024-07/2029). Total Cost: $393,974.
3. Choi, W (PI). “Interpretable Predictive Models for Radiation Therapy.” Sidney Kimmel Comprehensive Cancer Center at Thomas Jefferson University Hospital (01/2022-Present).

Past

1. Wilson, L (PI), Choi, W (Co-I). Microdosimetry for Prostate Cancer Radiotheranostics. Philadelphia Prostate Cancer Biome Project (02/2024-02/2025).
2. Shelton, J (PI), Choi, W (Subaward PI). Constructing a facial analytic deep learning training set for detecting frustration in early readers. Propel Health Impact Grant (05/2023-04/2024). Total Cost: $200,000 (TJU: $25,000).
3. Wang, J (PI), Choi, W (Co-PI). Deep Learning-based Target Tracking and Assignment for Cooperative Swamp Defense. DoD Naval Engineering Education Consortium (NEEC), N00174-21-1-0011 (05/2021-05/2024). Total Cost: $361,161.
4. Shetty, S (PI), Choi, W (Co-PI). Cybersecurity Monitoring and Assurance Training Program for Safe and Secure Port Operations. Commonwealth Cybersecurity Initiative Grant (05/2021-08/2022). Total Cost: $150,000 (VSU: $32,070).
5. Choi, W (PI). “Acquisition of a GPU-Accelerated Deep-Learning Research Cluster.” DoD Research and Education Program for HBCU/MI Equipment/Instrumentation, W911NF2110280 (05/2021-05/2022). Total Cost: $588,244.
6. Choi, W (PI). “A Robust Human Action Recognition System using Multi-View Depth Videos for Safe and Reliable Human-Robot Interactions in a Mixed Reality Environment.” Innovation Fund, VSU & Commonwealth Center for Advanced Manufacturing (CCAM E-055) (07/2020-07/2021). Total Cost: $50,000.
7. Choi, W (PI). 3D Image Analysis for Computer Aided Diagnosis in Lung CT Image. Basic Science Research Program, National Research Foundation of Korea, NRF-2013R1A1-A2058113 (11/2013-07/2014). Total Cost: $50,000.
8. Choi, W (PI). A System for Computer Aided Detection of Pulmonary Disease. Start-up Seed Grant, Small and Medium Business Administration of Korea, 2013RS-03-00000-P-00173 (12/2013-12/2014). Total Cost: $90,000.

Wookjin Choi, PhD
Associate Professor & Computational Physicist
Department of Radiation Oncology
Sidney Kimmel Medical College & Comprehensive Cancer Center
Thomas Jefferson University
August 2026